Khushi Sharma

Project Portfolio

UBC SOLAR



Battery Mechanical Senior Designer - Airflow Specialist

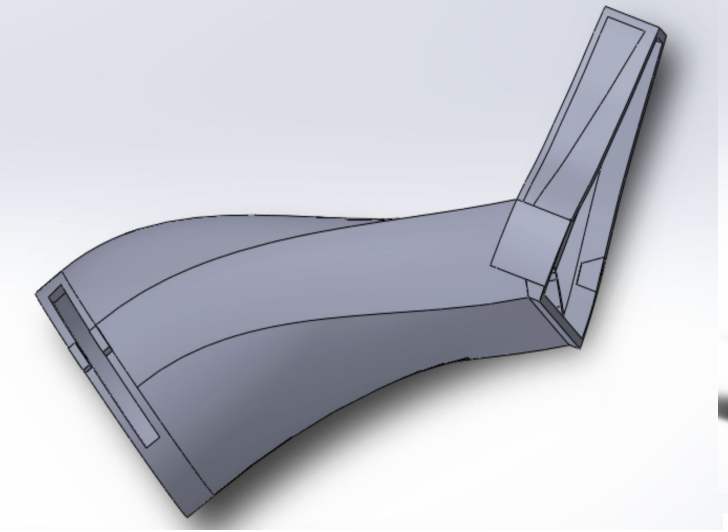
Joined: Sep 2024

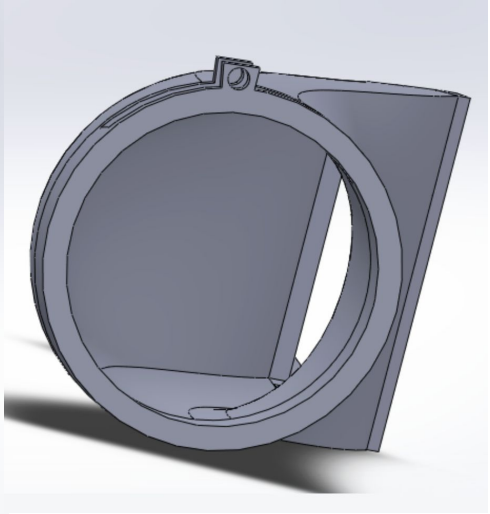
- Specialize in airflow and thermal management
- Lead CFD simulations
- **Overlook** all airflow projects
- Conduct **experimental tests**
- Make design decisions for our solar powered car operating in high temperature environments



Project 1: Intake to Aeroshell Duct

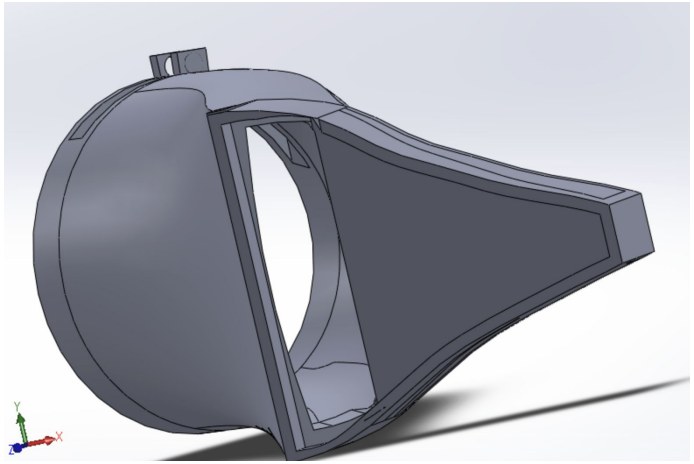
Jan 2026 - Present





Skills:

- SolidWorks
- Mechanical Design
- Thermal Airflow

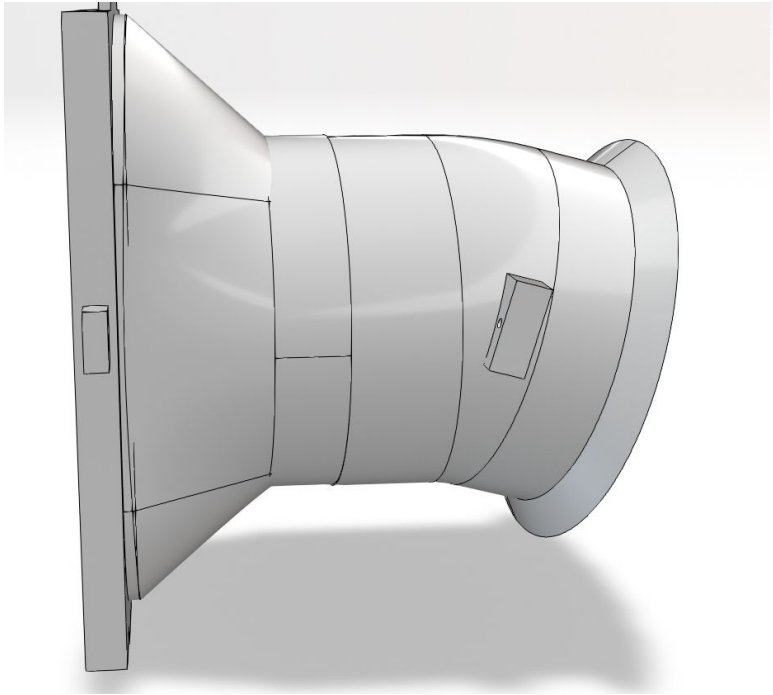


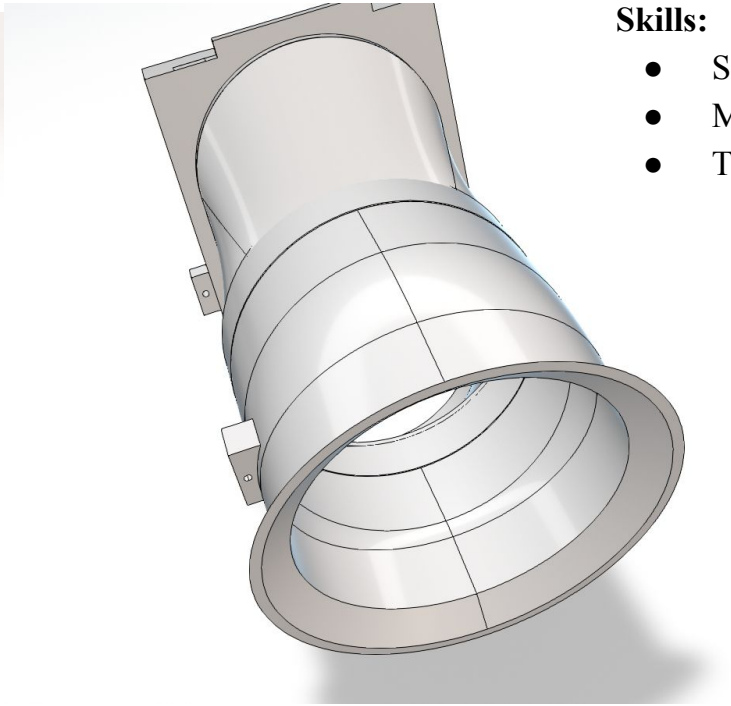
Key design features:

- NACA duct integrated into a diffuser for efficient battery cooling airflow.
- Smooth diffuser transition to minimize pressure losses.
- Mounting lip for quick installation into the aeroshell.
- Two-piece design for easy 3D printing and assembly.
- Integrated zip-tie mounts for waterproof mesh and battery impounding.
- Lightweight design to reduce overall vehicle mass.

Project 2: Battery Enclosure to Intake Duct

Jan 2026 - Present





Skills:

- SolidWorks
- Mechanical Design
- Thermal Airflow

Key design features:

- Smooth square-to-round transition and flared outlet optimize flow uniformity.
- Continuous internal surfaces prevent flow separation and pressure losses.
- Direct fitment connects the rectangular battery cutout to a circular flexible duct.
- Two-part split design uses flanges and bolts for quick 3D-print assembly.
- Integrated zip-tie holders fulfill competition battery impound requirements.

Project 3: Battery Pack Thermal Management & Airflow Optimization

Sep 2025 - Jan 2026

3.1 CFD Testing

Skills:

- SolidWorks Flow Simulation
- Solidworks
- Computational Fluid Dynamics (CFD)

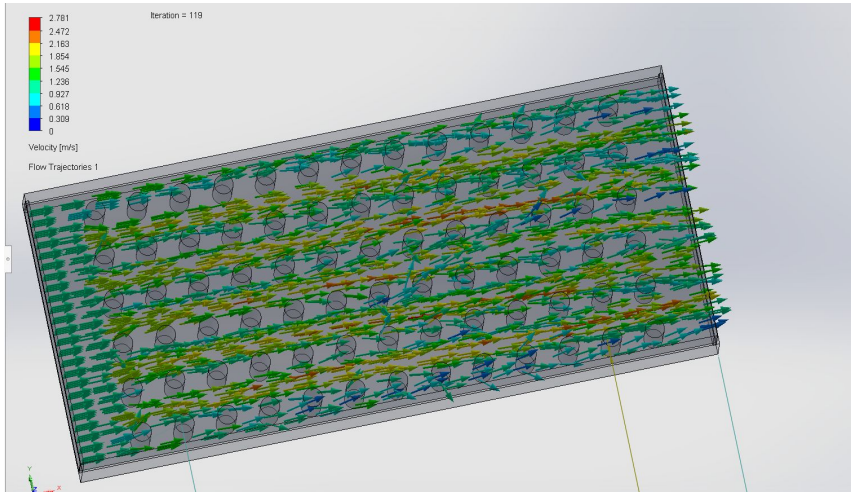


Figure 1: **Aligned** cell configuration

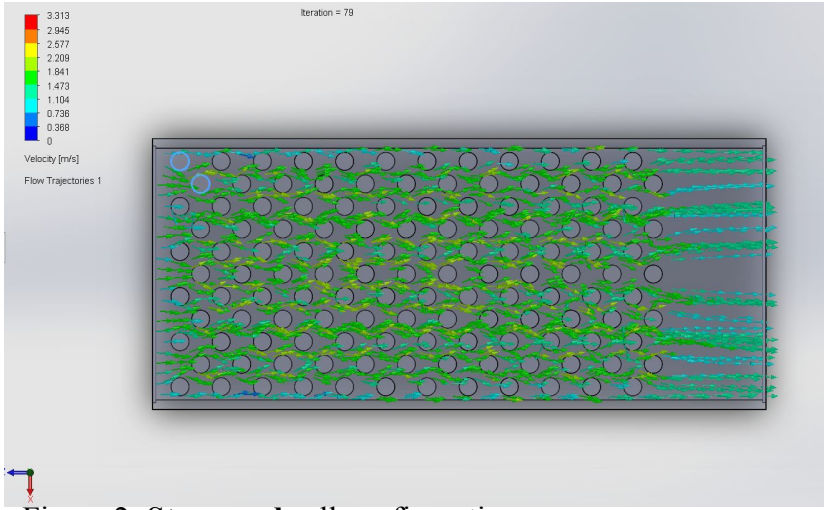


Figure 2: **Staggered** cell configuration

Optimizing Airflow through:

- Aligned vs. staggered cylindrical cell layouts
- Variations in X/Y cell spacing
- Resulting pressure drop and volumetric flow rate

Results: Staggered Cell Configuration with 24mm gap in X and 24mm gap in Y directions (Figure 2)

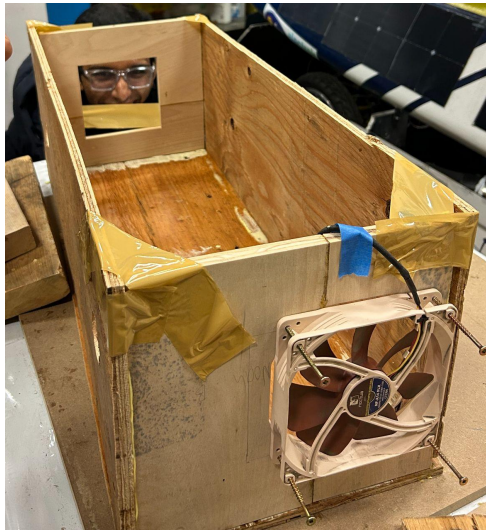
<u>Improved airflow by ~30% relative to the previous pack design</u>

3.2 Physical Testing (Experimental Validation)

Skills:

- Waterjet cutting
- 3D Printing
- Test plan creation
- airflow/pressure measurement using an anemometer

Designed and manufactured a wood airflow testing jig to experimentally evaluate battery pack configurations







Used a vane style anemometer to measure:

- Pressure drop across the mock pack
- Flow distribution through cell arrays
- Relative cooling effectiveness between layouts and duct designs

Results: In progress

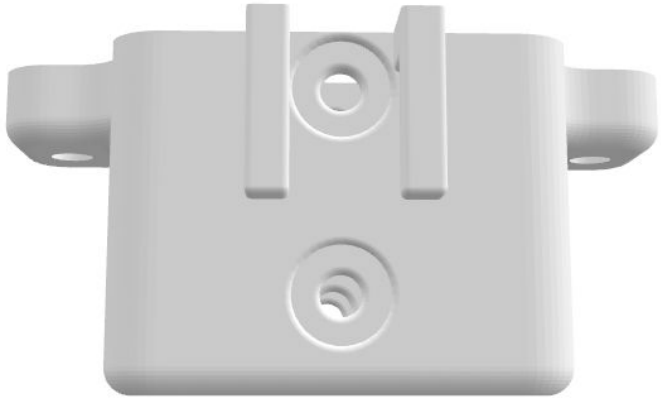
Project 4: Anderson Connector Isolation & Holder Design

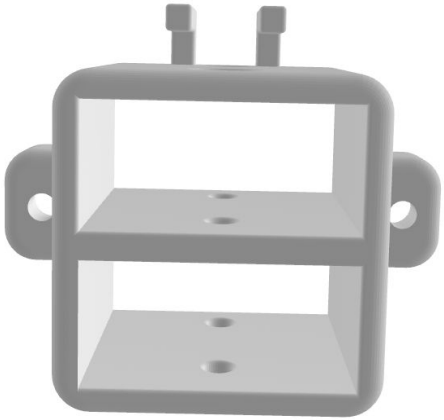
April 2025 - June 2025

Custom Anderson connector holder to securely isolate high-current connectors from the rest of the battery pack

Skills:

- SolidWorks
- Mechanical Design





Key design features:

- Geometry constrained side walls to prevent connector movement
- Integrated zip tie locking points for secondary retention
- Minimal fastener count to reduce weight while maintaining compliance

Project 5: Theoretical Airflow Simulation

Jan 2025 - Mar 2025

Theoretical airflow and heat dissipation study comparing stacked versus sheet-based battery module architectures for a next-generation battery pack

Tabulated Properties of Air			
T (K)	nu (m^2/s)	k (W/m K)	Pr
310	0.0000167	0.0271	0.70
320	0.00001766	0.0279	0.70
330	0.00001865	0.0286	0.70
340	0.00001966	0.0293	0.70
350	0.00002069	0.03	0.70
400	0.00002613	0.0335	0.69

values)		Cylinder Approximation Stack				Cylinder Approximation Sheets			
	k (W/m K)	Re_D	Nu_D	h (W/m^2 K)	Q(W)	Re_D	Nu_D	h (W/m^2 K)	Q(W)
53561475	0.027393131	1885.9563	8.64322	47.35298269765	4.69339	2828.9	11.954975	21.832280	316.5462276
53553770	0.027393837	1885.9563	8.64322	47.3541879992	5.16286	2828.8	11.954377	21.831750	287.7623114
53546066	0.027394544	1885.9563	8.64322	47.3553932999	5.63236	2828.6	11.953778	21.831220	263.7757154
53538361	0.027395250	1885.9563	8.64321	47.3565985999	6.10187	2828.4	11.953180	21.830690	243.4793657
53530656	0.027395956	1885.9563	8.64321	47.3578038990	6.57142	2828.2	11.952582	21.830160	226.0824952
53522951	0.027396662	1885.9563	8.64321	47.3590091973	7.04098	2828.1	11.951984	21.829630	211.0052081
53515246	0.027397369	1885.9563	8.64320	47.3602144948	7.51057	2827.9	11.951385	21.829101	197.8125824
53507541	0.027398075	1885.9563	8.64320	47.3614197914	7.98019	2827.7	11.950787	21.828571	186.1720310
53499836	0.027398781	1885.9563	8.64320	47.3626250873	8.44982	2827.5	11.950189	21.828042	175.8248746
53492131	0.027399487	1885.9563	8.64320	47.3638303823	8.91949	2827.4	11.949592	21.827512	166.5668931
53484426	0.027400193	1885.9563	8.64319	47.3650356766	9.38917	2827.2	11.948994	21.826983	158.2347103
53476721	0.027400900	1885.9563	8.64319	47.3662409700	9.85888	2827.0	11.948396	21.826453	150.6960691
53469016	0.027401606	1885.9563	8.64319	47.3674462626	10.32861	2826.8	11.947798	21.825924	143.8427593
53461311	0.027402312	1885.9563	8.64319	47.3686515544	10.79837	2826.7	11.947201	21.825395	137.5853900

Skills:

- First-principles thermal modeling
- Analysis → Strategic design decision

[Link To Simulation]

Methodology:

- First-principles thermal modeling incorporating pack geometry
- Approximate race-condition heat loads and airflow assumptions
- Comparison of effective heat rejection between layouts

Results: Sheets-based module layout experienced **70x higher** heat accumulation therefore indicating superior airflow in the stack module design configuration

The analysis directly informed design decisions, leading the team to select the stacked module architecture for the 2027 competition battery pack

Project 6: Carbon Fiber Battery Enclosure Fabrication

Jan 2025 - Mar 2025

Designed and manufactured carbon fiber composite panels for the battery pack enclosure using an epoxy resin system and vacuum bagging.





Skills:

- Composite Design and Manufacturing
- Epoxy Resin processing
- Vacuum Bagging

Process Highlights

- Mold preparation and multi-ply fiber layup
- Resin infusion and controlled curing
- Design optimization for vibration resistance and structural stiffness

The resulting enclosure provided robust mechanical protection while achieving a high stiffness-to-weight ratio, reducing overall battery pack mass compared to metal alternatives and improving vehicle energy efficiency

Personal Projects

[(Link to Project Video)]

[(Link to Github Repository)]

Skills:

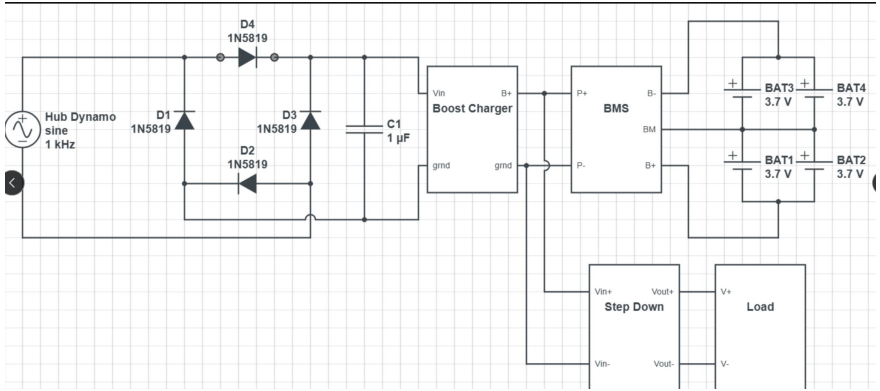
Electrical System Design: AC/DC conversion, battery integration, and voltage regulation

CAD & Prototyping: SolidWorks modeling and 3D-printed component fabrication

Testing & Iteration: Performance testing, efficiency analysis, and design optimization

ReCycle: Bicycle-Mounted Power Generation and Charging System

ReCycle converts mechanical energy from wheel rotation into electrical energy, allowing riders to store charge in a custom battery pack and power devices on long-distance cycling trips. Received first place among second-year capstone projects by audience choice.



Key Design Features:

- Integrated a **hub dynamo generator** with the bicycle wheel to convert rotational mechanical energy into electrical power during riding.
- Designed an AC-to-DC conversion circuit using a **bridge rectifier and filtering capacitor** to create stable power for storage.
- Developed a **2S2P lithium-ion battery pack** integrated into the bicycle frame to store harvested energy.
- Designed and fabricated a **custom 3D-printed phone mount and protective enclosure** using SolidWorks to securely house system components.
- Optimized the system integration to minimize rider impact, achieving only a **+1.76 minute time penalty over 20 km** while enabling on-the-go device charging.

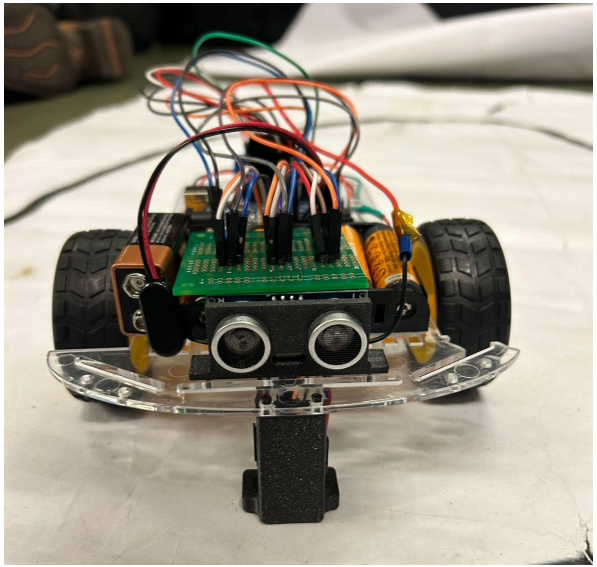
[(Link to Project Video)]

Skills:

- Embedded systems programming
- Mechatronic system integration
- Control tuning and iterative prototyping

Line Following Robot Project

Designed, built and programmed a two-wheel differential-drive robot using an Arduino Uno and L298N to autonomously follow a predefined line



Key Design Features:

- Implemented real-time sensor-based control with PWM motor modulation to correct path deviation and enable stable tracking
- Integrated mechanical assembly, embedded electronics, and firmware development
- Iteratively tuned control parameters to improve reliability across varying track conditions

[Article link]
[GitHub Repository]

Skills:

- Deep Learning Model
- Natural Language Processing (NLP)
- Machine Learning

Machine Translation Transformer

Built a transformer-based machine translation model to translate educational and administrative materials, including textbooks and admission forms.



Photo taken mid lesson at Resources for Youth in India

Key design features:

- Used attention mechanisms to improve translation accuracy through structured and unstructured text to enhanced accessibility to educational resources
- Created specifically for Resources for Youth in India organization [that I was leading at the time]

Skills:

- Chemical Engineering problem solving
- Adsorption-based water treatment design
- Pitching to Investors

[Project Memo]

[Presentation pitch]

Clean Water in Haiti - Ammonium Removal System

Designed a water treatment solution targeting ammonium ion removal using adsorption-based techniques to address metal contamination in Haiti's drinking water



This project received funding from the 1517 Fund and was waitlisted for implementation by Haiti's water board following technical review

Skills:

- Innovation Design and Validation
- Scenario based problem solving
- Pitching to executives

[(Stand alone deck)]

Walmart Canada Consulting Challenge

Developed a three-part systems-level strategy proposing operational, technological, and customer-experience transformations to position Walmart for retail engagement in 2030



The deck was selected among the top 7 globally and presented to the CEO of Walmart and the retail innovation team. This project was then worked on further during my time interning at Walmart Canada's innovation hub.

Skills:

- Scientific Writing and publication
- Analysis of emerging technologies

[(Link to Paper)]

Cellular Agriculture Review Paper

Authored a technical review paper analyzing cultured meat production from a cellular agriculture perspective, covering bioprocessing challenges, scalability constraints, and commercialization pathways.



This review paper was peer-reviewed by Kristopher Gasteratos [founder of the Cellular Agriculture Society]

Skills:

- Public speaking

[(Link to Talk)]

TEDx Talk - Redefining Learning: The Disruptive Power of AI in Education

Delivered a TEDx talk exploring how artificial intelligence can transform education through accessibility, personalization, and adaptive learning systems.

